Mona Pouresmaeil

MSc Computer Engineering – AI and Data Analytics

Turin, Piedmont, Italy

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Professional Summary

AI Engineer specialized in **NLP**, **RAG**, and **agentic workflows** with a proven track record of building production-grade LLM applications. Currently finalizing a Master's thesis at **Brain Technologies** focused on advanced document validation frameworks, with **full immediate availability** for full-time roles.

Co-author of the **SemEval 2026 Task 1** system that ranked **2nd internationally**. Expert in developing fully **custom AI pipelines from scratch** in pure Python, leveraging **LangGraph**, **NLP** techniques, and hybrid search to solve complex industrial problems.

Actively seeking **AI/ML Engineer** or **LLM Engineer** positions to apply advanced expertise in **NLP** and information retrieval systems immediately.

Education

Politecnico di Torino — *Turin, Italy* — 2023 – July 2026

- **MSc in Computer Engineering** – Artificial Intelligence & Data Analytics Track

Shariaty Technical and Vocational College — *Tehran, Iran*

- **BSc in Computer Technology Software Engineering** — 2020 – 2022
- **Associate Degree in Computer Software Engineering** — 2018 – 2020

Experience

AI Engineer Intern — *Turin, Italy (On-site)* — Jan 2026 – Present

Brain Technologies

- Finalizing Master's thesis: developed a fully custom AI framework for automatic search, validation, and verification of documentary content using **RAG**, **advanced NLP techniques**, and **LLM-based pipelines** built entirely from scratch
- Designed and implemented the complete RAG pipeline manually, including **custom chunking strategies**, embedding generation, **hybrid search (vector + keyword)**, **confidence scoring** for **hallucination mitigation**, and compliance features
- Gained hands-on experience building **production-oriented LLM applications** and **information retrieval** systems from the ground up in pure **Python**

Backend Developer Intern — *Tehran, Iran (Remote)* — Mar 2020 – Aug 2020

Rastava Academy

- Developed scalable backend services using **Python and Django**
- Collaborated in **Agile** teams with **Git/GitHub** for version control and code reviews
- Designed and optimized **database schemas** for data integrity and efficient retrieval

Software Engineering Intern — *Tehran, Iran (On-site)* — Sep 2019 – Feb 2020

Kasra Company

- Customized enterprise software solutions for over 500 clients according to specific business needs and workflows
- Optimized **SQL queries** to improve database performance and reduce latency
- Participated actively in **Agile development cycles**

Key Projects

SemEval 2026 Task 1 – Humor Generation

- Co-developed humor generation system that ranked **2nd internationally**
- Implemented **LLM-based generation** and evaluation workflow leveraging **RAG** and **preference optimization** techniques
- **Tech:** Python, NLP, RAG, **DPO**, **Large Language Models**

RAG Document Validation Framework

- Built AI system for automatic search, validation, and verification of documentary content using **RAG** and **LangGraph**
- Implemented ingestion for PDF/DOCX/PPTX, local **Llama 3** Q&A via **Ollama**, and answer/requirement verification for compliance workflows
- **Tech:** Python, **LangChain**, **LangGraph**, Ollama (Llama 3), **ChromaDB**, **FAISS**

Agentic Job Tracker 2026

- Built AI-powered tool for Italian job postings with auto-translation, **field extraction**, tax estimator, and **agentic chat interface**
- Extracted key fields (company, location, RAL, tech stack) and generated **structured summaries** to speed up application triage
- **Tech:** Python 3.13, **Streamlit**, **Llama 3.3-70B**, **Hugging Face**, **Agentic AI**

Fake News Detection System

- Created complete **NLP pipeline** with **TF-IDF** and classification models (**SVM**, **Logistic Regression**, **LSTM**)
- Compared models and tuned preprocessing/features to improve robustness across datasets
- **Tech:** Python, **scikit-learn**, NLTK, LSTM

Deep Learning Handwritten Digit Classifier

- Implemented advanced **CNN** in **PyTorch** achieving **99%+ accuracy** on MNIST
- Improved generalization using **batch normalization**, **dropout**, and data augmentation during training
- **Tech:** PyTorch, CNN, Batch Normalization, Dropout

Skills

- **AI & Machine Learning: Machine Learning**, **Deep Learning**, **NLP**, **LLMs**, **Retrieval-Augmented Generation (RAG)**, **Agentic AI**, Feature Engineering, **Hallucination Mitigation**, **Prompt Engineering**, Information Retrieval, Hybrid Search, Custom Chunking & Embedding
- **RAG Engineering:** From-scratch pipeline development in pure Python, custom chunking, embedding workflows, hybrid retrieval (vector + keyword), and confidence scoring
- **Frameworks & Libraries: Python**, **PyTorch**, **scikit-learn**, **LangChain**, **LangGraph**, **Hugging Face**, TensorFlow, XGBoost, Pandas, NumPy, spaCy, NLTK, **Ragas**, **Qdrant**, **Docling**, **Unsloth**, **ChromaDB**, **FAISS**
- **Software Engineering & Tools:** Django, SQL, Query Optimization, React, **REST APIs**, Git/GitHub, **Docker**, **AWS**, Streamlit, Flask, Agile, Jupyter, Google Colab

Certifications

- **AI Agents Intensive Course with Google** (2025)
- **RAG for Professionals with LangGraph, Python and OpenAI** – Udemy (Dec 2025)
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning – DeepLearning.AI (Nov 2025)
- GenAI Intensive Program – Kaggle

Languages

- **English** — C1 (Fluent)
- **Italian** — A2-B1 (actively improving)